

Relaxor-like ferroelectric response and achiral–chiral switching in an amorphous molecular solid

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 Check for updatesHanyu Liu¹✉, Yuqin Xiong¹, Ti Xie², Cheng Gong² & Shenqiang Ren¹✉

Organic ferroelectrics offer a molecular platform for integrating polarization dynamics with molecular design, photoresponsiveness and chiroptical functions, including light-triggered achiral-to-chiral transitions, yet these responses remain strongly shaped by crystalline packing in the solid state. Here, we vitrify a salicylideneaniline derivative that undergoes UV-triggered achiral-to-chiral conversion and show that structural disorder enables coupled chiroptical and ferroelectric responses in an amorphous molecular solid. The vitrified state loses long-range order while retaining spatially heterogeneous local polar responses, giving a coercive field of $\sim 5.7 \text{ kV cm}^{-1}$ and a maximum switchable polarization of $\sim 26 \text{ } \mu\text{C cm}^{-2}$. It also exhibits over an order of magnitude faster recovery from the chiral photoactivated state than the crystalline counterpart. Moreover, circularly polarized light produces helicity-dependent dielectric and thermal responses with mirror-symmetric behavior between the (S)- and (R)-enantiomers. These results connect photoinduced symmetry switching with relaxor-like dipolar dynamics in a single-component amorphous molecular ferroelectric.

Ferroelectric materials have long served as the cornerstone of polarization-based electronics and sensing technologies, yet traditional systems are typically limited by rigid crystalline symmetry and restricted tunability. Organic ferroelectrics are redefining this landscape by integrating molecular design, optical activity, and polarization dynamics within a single, flexible framework^{1–4}. Unlike their inorganic counterparts, these materials inherently support photoresponsive and chiroptical behaviors, enabling dynamic modulation of polarization and symmetry under external stimuli such as light or electric fields—an appealing feature for wearable neuromorphic circuits^{5–7} and light-controlled sensors^{8–10}. Within this class, photoresponsive crystalline ferroelectrics have demonstrated that light can modulate polarization, optical absorption, and molecular chirality through well-defined structural pathways^{7,11–14}. Particularly, achiral-to-chiral transitions, where molecular asymmetry and optical activity response can be reversibly introduced or erased through external activation, offers a route to control symmetry at the molecular scale. In

crystalline molecular solids, however, such responses are intrinsically constrained by long-range packing order, fixed lattice symmetry, and relatively slow collective structural relaxation. By contrast, amorphous molecular solids provide a fundamentally different physical environment: they lack translational symmetry but retain local molecular order and heterogeneous dipolar configurations. These features may enable polarization dynamics that are inaccessible in crystalline counterparts, including broad relaxation, frequency dispersion, and rapid structural reconfiguration. Despite this potential, whether switchable polarization and relaxor-like dipolar behavior can emerge in a fully amorphous, single-component molecular solid remains largely unexplored. Relaxor ferroelectricity is particularly relevant in this context^{15–21}. In classical relaxors, nanoscale polar heterogeneity, distributed local environments, and dynamic dipolar freezing give rise to diffuse dielectric anomalies and strong frequency-dependent polarization responses. To date, such behavior has been realized mainly in crystalline or semicrystalline inorganic oxides, polymers, and

¹Department of Materials Science and Engineering, University of Maryland, College Park, MD, US. ²Department of Electrical and Computer Engineering, University of Maryland, College Park, MD, US. ✉e-mail: hanyuliu@umd.edu; sren@umd.edu

composite systems, where disorder is introduced through chemical substitution, random fields, or mesoscale phase separation. Extending relaxor-like ferroelectricity to a pure amorphous molecular solid would therefore represent a distinct mechanism: one in which dipolar disorder arises not from compositional complexity but from vitrified molecular packing and locally frustrated polar configurations.

Here, we investigate this possibility using a salicylideneaniline derivative, N-3,5-di-*tert*-butylsalicylidene-1-4-bromophenylethylamine (SA-Ph-Br). The two bulky *tert*-butyl substituents in the molecular introduce substantial steric hindrance, frustrate efficient close packing and suppress long-range crystallization during rapid vitrification, thereby favoring the formation of a kinetically trapped amorphous phase. We show that amorphous SA-Ph-Br exhibits switchable polarization and relaxor-like dielectric behavior in a single-component molecular glass. Upon rapid vitrification, the amorphous (S/R)-SA-Ph-Br loses long-range periodicity but retains local polar order, exhibiting a coercive field of -5.7 kV cm^{-1} (substantially lower than that reported, -40 kV cm^{-1} for the crystalline phase²²), a maximum switchable polarization ($-26 \mu\text{C cm}^{-2}$), and pronounced frequency-dependent dielectric relaxation. It has reported that salicylideneaniline derivative undergoes a UV-induced achiral-to-chiral transition arising from enol-keto tautomerism, in which intramolecular proton transfer drives reversible changes in molecular geometry and dipole moment^{22–28}. This transition modulates both visible optical absorption and the electric dipole arrangement, thereby coupling optical activity and ferroelectric polarization. Distinct from previous studies of crystalline photoresponsive SA-Ph-Br and related molecular ferroelectrics, we establish how amorphization qualitatively reshapes the coupled chiroptical and ferroelectric responses. In the amorphous state, the absence of rigid lattice constraints enables much faster recovery from the UV-activated chiral state to the ground achiral state, occurring within $\sim 1 \text{ min}$ at room temperature, compared with more than 10 min in the crystalline phase. The relaxation time can be further extended to $\sim 50 \text{ min}$ at 5°C , indicating that the photoinduced chiral state is governed by thermally tunable molecular mobility in the amorphous matrix. Furthermore, the photoferroelectric response is demonstrated to be intrinsically chiral, presenting helicity-dependent dielectric function, thermal conductivity and saturation polarization. Collectively, this work establishes amorphous molecular solids as a previously unexplored platform for relaxor ferroelectricity and reveals a general strategy for coupling photoinduced chirality, dipolar disorder and molecular vitrification to achieve reconfigurable ferroic behavior. Collectively, this work establishes amorphous molecular solids as a previously unexplored platform for relaxor-like ferroelectricity and differentiates the amorphous SA-Ph-Br system from crystalline photoresponsive ferroelectrics by revealing disorder-enabled, rapidly reconfigurable, and helicity-sensitive ferroic behavior beyond crystalline order.

Results

Figure 1 illustrates the correlation between crystalline (slow-cooled) and amorphous (vitrified) phases of the molecular ferroelectric SA-Ph-Br, and their connection to the reversible achiral-to-chiral transition driven by light-activated enol-keto tautomerization. As shown in Fig. 1a, the structural order of SA-Ph-Br can be reversibly tuned through thermal processing, yielding a tunable relaxor-like ferroelectric system that couples structural, electronic, and chiroptical properties. Slow cooling of molten SA-Ph-Br results in a ferroelectric crystalline solid with long-range structural order, whereas rapid cooling induces vitrification, forming an amorphous molecular polar glass exhibiting relaxor-like dipolar dynamics. (Fig. 1b) During this quenching process, incomplete molecular relaxation traps the system in a kinetically arrested state with only short-range structural correlations. Upon annealing near the cold-crystallization temperature, the amorphous phase can transform into a more ordered crystalline structure,

highlighting the metastable nature of the vitrified state. The amorphous material exhibits either a glassy or rubbery character depending on the temperature relative to its glass transition temperature (T_g). The loss of long-range periodicity disrupts extended dipolar alignment, giving rise to frequency-dispersive dipolar behavior associated with dynamic heterogeneous local polar response^{29–33}. Crucially, this structural disorder also modulates the achiral–chiral photoresponse of SA-Ph-Br (Fig. 1c). Although both amorphous and crystalline phases undergo an enol–keto transition, the disrupted hydrogen-bond network in the amorphous state facilitates rapid and reversible light-driven switching between achiral (vis-inactive) and chiral (vis-active) states³⁴. Here, we demonstrate that the amorphous phase presents ~ 10 times quicker recovery from chiral to achiral states compared to crystalline phase. Moreover, we demonstrated that the intrinsic chirality of SA-Ph-Br allows its physical properties to be modulated by the handedness of incident circularly polarized light (CPL).

Figures 2a, b compare the optical appearance and the surface texture of crystalline and amorphous SA-Ph-Br films. As shown in the optical photos, the amorphous film exhibits a light-yellow hue with a uniform, glass-like surface, while the crystalline film, though similarly colored, shows distinct grain boundaries characteristic of polycrystalline morphology. The corresponding SEM images confirm that the amorphous film has a significantly smoother surface than the crystalline counterpart, consistent with its glassy appearance. Particularly, we acquired the AFM images at the crystalline–amorphous boundary highlight the contrast of surface morphology, where the amorphous area shows great flatness compared to the crystalline area, potentially benefiting contacts in device manufacturing. The chiroptical and optical absorption properties of the (S/R)-amorphous films were investigated using circular dichroism (CD) and UV–vis spectroscopy (Fig. 2c). Both S- and R-enantiomeric amorphous films display symmetric CD spectra of opposite sign, confirming their enantiomeric nature. In their pristine state, the amorphous films show negligible CD response and minimal visible absorption above 400 nm . Upon UV irradiation, however, pronounced CD and absorption bands appear in the visible region, indicating the emergence of photoinduced chiral electronic states. UV excitation triggers an enol–keto intramolecular proton transfer, activating new visible optical transitions²². The corresponding chiroptical and optical properties of the crystalline films were also investigated and presented in Fig. 2d. Notably, the induced visible CD band exhibits an opposite sign relative to the UV peak—distinct from the behavior of crystalline films, potentially associated with altered hydrogen-bond configurations in the amorphous phase. Structural differences between the amorphous and crystalline states are further revealed by X-ray diffraction (XRD) (Fig. 2e). The amorphous film exhibits a featureless diffraction pattern, confirming the absence of long-range order, whereas the crystalline film displays sharp diffraction peaks consistent with reported patterns²². We acquired the TGA curve of prepared SA-Ph-Br (Supplementary Fig. 1) and FTIR and Raman spectra (Supplementary Fig. 2) of amorphous and crystalline films. The result of crystalline phase agrees well with previous literature while result of amorphous phase shows systematic shifts in vibrational bands, reflecting modifications in molecular packing and hydrogen-bonding interactions. Particularly, the less packed stacking of molecules in amorphous phase leads to the general redshifts of FTIR peaks associated with *tert*-butyl groups showing less constraint in the amorphous phase. 1590 cm^{-1} peak assigned to azomethine C=N stretching also presents a clear red shifting from the 1600 cm^{-1} in crystalline phase. Such shifting may be associated with modified intramolecular hydrogen-bond environments in the amorphous phase, which is the result of a weaker intermolecular hydrogen bonding^{27,35–38}. The hydrogen bond plays an important role in maintaining lifetime of the keto state (vis-active chiral state), and therefore it lays the fundamental ground for tunability of chiral–achiral transition. Thermal behavior of the transition between crystalline and

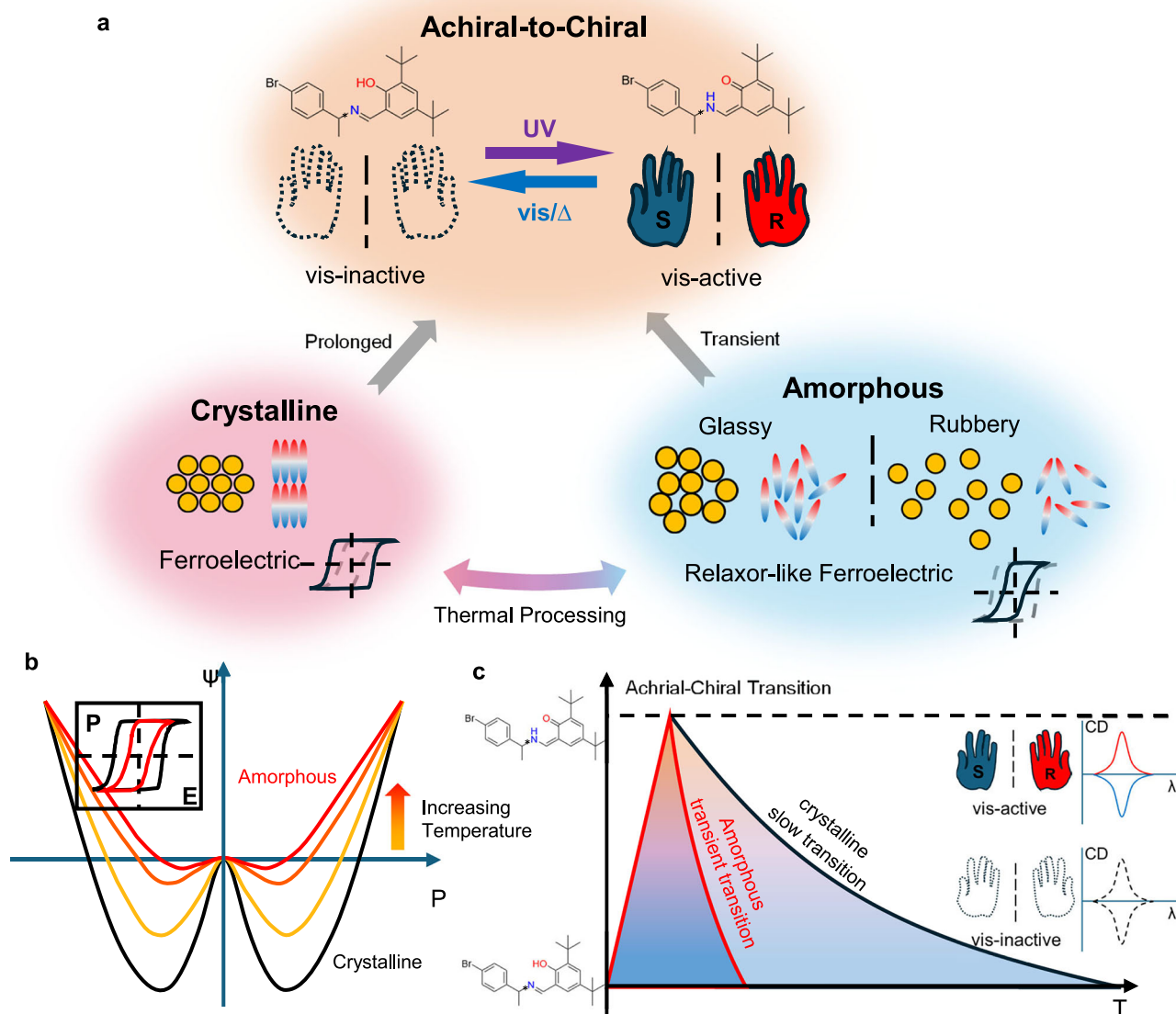


Fig. 1 | Coupled crystalline–amorphous and achiral–chiral transitions underpin a reconfigurable chiral molecular relaxor ferroelectric platform. a Interplay of crystalline–amorphous structural transition and achiral–chiral mechanism enables

reconfigurable chiral molecular ferroelectric platform, **b** Landau free energy versus polarization for amorphous and crystalline phases. **c** Chiral to achiral transition curve for crystalline and amorphous phases.

amorphous phases was characterized by differential scanning calorimetry (DSC) (Fig. 2f), performed at a heating/cooling rate of $10^\circ\text{C min}^{-1}$. Starting from the crystalline phase, during the first heating cycle, two ferroelectric transition peaks appear near -32°C and 62°C , which were assigned as the ferroelectric transitions in the ferroelectric crystalline form²², followed by an endothermic melting peak at -107°C . Upon cooling, no crystallization exotherm is observed due to rapid quenching, while a glass-transition step near 20°C emerges in both cooling and subsequent heating cycles. An additional endothermic feature around -25°C upon cooling corresponds to the freezing of dipolar motion. The reversible thermal transformation from amorphous states to crystalline states can be achieved with the cold crystallization, the thermal feature of which was clearly identified at a slower heating rate (2°C min^{-1}). (Supplementary Fig. 3) Lack of long-range structural order in vitrification-induced amorphous SA-Ph-Br solids were chosen for further investigation owing to their short-range structural order, which facilitates a facile achiral-to-chiral transition and reduced ferroelectric coercivity. XRD analysis of the aged

amorphous sample stored under ambient conditions for 5 months shows that the material remains predominantly amorphous, without the appearance of a complete set of sharp Bragg peaks characteristic of the crystalline phase (Supplementary Fig. 4). The weak residual diffraction features are attributed to local or medium-range molecular ordering, likely associated with preferential packing along the molecular long axis, rather than full crystallization. The slight evolution of the broad amorphous halo in the $20\text{--}25^\circ$ region suggests slow structural relaxation within the amorphous matrix, consistent with physical aging, reduced free volume, and gradual local densification. Thus, the amorphous phase is kinetically stable against crystallization over this time scale while undergoing modest structural relaxation within the disordered state.

The amorphous state enables structural-disorder-mediated tuning of the photoinduced chiral-to-achiral transformation in SA-Ph-Br (Fig. 3a). FTIR analysis has indicated that amorphization weakens and broadens the hydrogen-bonding environment associated with the salicylideneaniline proton-transfer unit. This weakened hydrogen

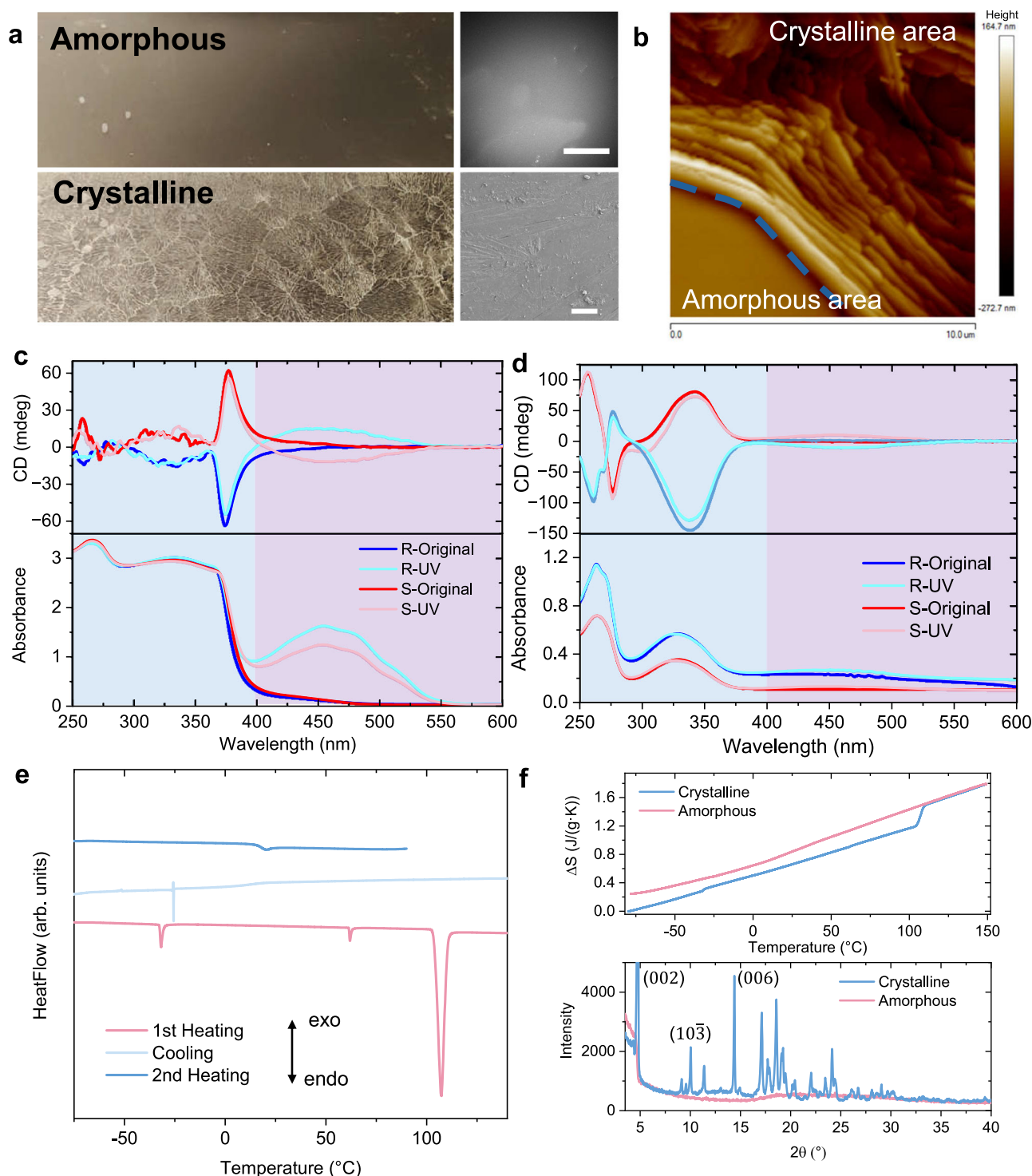


Fig. 2 | Characterization and comparison of amorphous and crystalline SA-Ph-Br. **a** optical images of crystalline and amorphous films. Insets: SEM images of the crystalline and amorphous films, Scale bar: 100 μm . **b** AFM image characterizing the surface morphology of crystalline and amorphous area. **c** CD and absorbance

spectra of R- and S- amorphous film with and without UV irradiation. **d** CD and absorbance spectra of R- and S- crystalline film with and without UV irradiation. **e** DSC showing the crystalline-amorphous transition. **f** Comparison of entropy and XRD of the crystalline and amorphous films.

bonding reduces stabilization of the proton-transferred keto configuration and weakens the intermolecular interaction that traps the UV-induced chiral state. Consistent with salicylideneaniline-type photochromic systems, the lifetime of the keto photoproduct is governed by hydrogen-bond strength and geometry, which controls both keto-state stability and the barrier for reverse proton transfer^{23,39}. Thus, weaker hydrogen-bond stabilization in the amorphous phase is expected to raise the relative free energy of the keto-state minimum

and/or lower the recovery barrier, resulting in a shorter keto-state lifetime. The recovery kinetics are further regulated by the glass transition of the amorphous matrix. Near or above the glass-transition region, increased molecular mobility and free-volume redistribution promote conformational relaxation and reverse proton transfer, accelerating the keto-to-enol recovery. Below the glass transition, however, the dynamically arrested amorphous network restricts local molecular rearrangement, thereby prolonging the lifetime of the

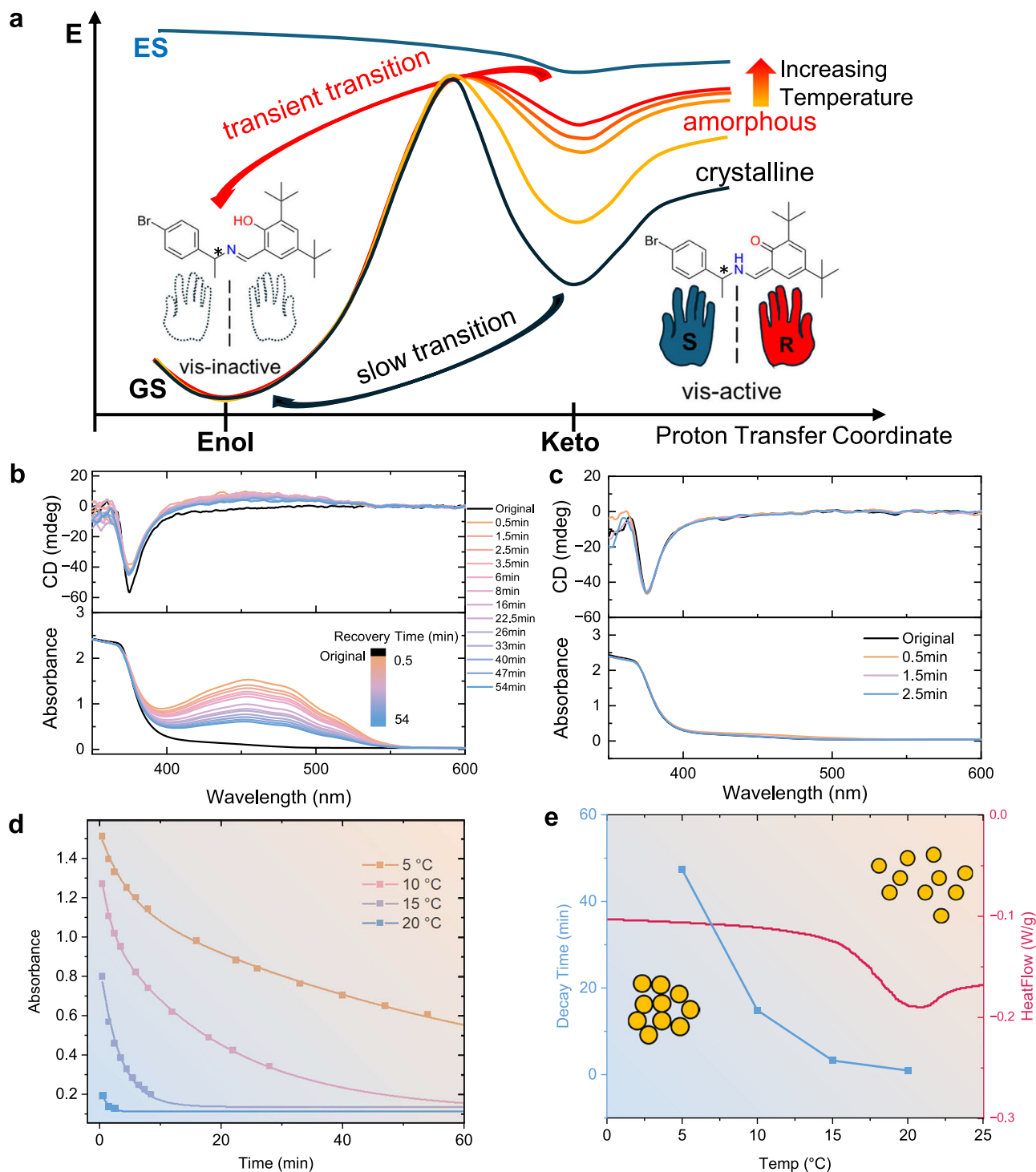


Fig. 3 | Chiral-achiral transition dynamics based on the intramolecular molecular transition tunable by temperature. **a** phenomenological energy-landscape schematic describing the photoinduced keto/chiral state and its relaxation to the enol/achiral ground state. **b** In-situ CD study studying chiral-achiral transition at

5 °C. **c** In-situ CD study studying chiral-achiral transition at 20 °C. **d** Extracted absorbance of 450 nm for different temperature. **e** Variance of recovery time compared to the thermal analysis of the glass transition.

photoinduced keto/chiral state. To verify this picture, we performed in-situ circular dichroism (CD) and UV-vis absorbance measurements on the R-amorphous film at various temperatures (5 °C, 10 °C, 15 °C, and 20 °C; Fig. 3b, c and Supplementary Fig. 5a, b). After UV activation, the sample was kept in the dark while sequential CD spectra were recorded at timed intervals. Both the CD intensity and visible absorbance gradually decayed to their initial achiral (vis-inactive) state, reflecting relaxation of the keto form back to the enol tautomer via

reverse proton transfer. For quantitative analysis, the in-situ absorbance of 450 nm at various temperatures were extracted and fitted by exponential decay function, $A = A_0 e^{-t/\tau}$ in Fig. 3d and the relaxation time, τ from fitting were plotted in Fig. 3d. The recovery slows markedly with decreasing temperature, indicating a thermally activated process involving intramolecular proton-transfer and hydrogen-bond rearrangement, which corresponds to the glass transition near 20 °C. (Fig. 3e) Note that at 5 °C, the measured data slight deviates from the

exponential fitting curve. We believe that this is due to the inhomogeneous distribution of relaxation time because of inhomogeneous size and strength distribution of the formed intramolecular hydrogen bond network. Near room temperature, the relaxation process occurs too rapidly to be fully resolved with our in-situ setup, whereas crystalline films exhibit much slower decay dynamics (Supplementary Fig. 6). This contrast highlights that structural disorder strongly modify the relaxation kinetics of the photoinduced state, thereby enabling rapid and reversible optical switching.

The ferroelectric behavior of the S-amorphous SA-Ph-Br film was characterized by polarization–electric field (P–E) hysteresis loops and current density–electric field (J–E) loops measured at room temperature and 200 mHz (Fig. 4a). The corresponding P–E and J–E loops of the R-amorphous SA-Ph-Br film were also collected and are presented in Supplementary Fig. 7. Frequency-dependent and voltage-dependent P–E loops were further measured and are shown in Supplementary Figs. 8 and 9, respectively. The film exhibits a clear hysteretic loop with a saturation polarization of $-1 \mu\text{C cm}^{-2}$. The corresponding coercive field (-5.7 kV cm^{-1}) is markedly smaller than that of the crystalline counterpart (-40 kV cm^{-1}), indicating enhanced dipolar mobility and a lower energy barrier for polarization reversal in the amorphous state. To examine local ferroelectric order, piezoresponse force microscopy (PFM) measurements were performed (Fig. 4b). PFM measurements reveal spatially heterogeneous electromechanical contrast with characteristic submicron-to-micron-scale features, supporting the presence of locally heterogeneous polar response in the amorphous phase. The corresponding topographic image (Supplementary Fig. 10) reveals no correlated morphological features, ruling out topography-induced artifacts and validating the PFM signal as intrinsically ferroelectric in origin. To examine whether electric-field poling induces permanent structural modification or field-assisted crystallization, we performed FTIR comparison before and after poling. (Supplementary Fig. 11) The poled film retains the major vibrational fingerprints of the pristine film, indicating that the applied electric field does not induce detectable chemical transformation or crystallization. The Nevertheless, subtle intensity and lineshape variations are observed in the $650\text{--}900 \text{ cm}^{-1}$ region, which is dominated by aromatic C–H out-of-plane bending, substituted-ring deformation, and coupled framework modes of the salicylideneaniline unit. These modes are known to be sensitive to the molecular environment and ring-twisting conformation of salicylideneaniline derivatives⁴⁰. Therefore, the spectral changes after poling are consistent with field-induced redistribution of local molecular orientation and ring-twisting conformations, possibly involving a subtle change in the distribution of inter-ring dihedral angles. The dielectric response of the amorphous film as a function of frequency and temperature (Fig. 4c, d) shows both real and imaginary components of permittivity featuring peaks that shift to higher temperatures with increasing frequency (40 Hz–1 kHz), further revealing relaxor-like dipolar dynamics. Extracting the peak positions of the real and imaginary part of the dielectric permittivity, a clear upshifting can be seen as the frequency increases (Fig. 4e). The Vogel-Fulcher fitting shows a reasonable agreement, suggesting glass-like slowing-down of dipolar relaxation processes.

Temperature-dependent PUND (Positive-Up-Negative-Down) measurements were conducted to probe dynamic polarization switching (Fig. 4f). The extracted switchable polarization (ΔP and $-\Delta P$) increases with temperature at low temperatures and reaches a maximum near -3.5°C . The reduction of switchable polarization at lower temperature reflects kinetic freezing of locally correlated dipolar configurations within the disordered matrix, consistent with relaxor-like dielectric dynamics. At higher temperatures, both ΔP and $-\Delta P$ decrease sharply as enhanced thermal motion and accelerated dipolar relaxation weaken cooperative polarization. The maximum ΔP value of $-26 \mu\text{C cm}^{-2}$ notably exceeds that of the crystalline phase, underscoring the strong local dipolar correlations and enhanced

polarizability within the amorphous ferroelectric network. Temperature-dependent I–V measurements also support the temperature-dependent ferroelectric response and indicate that the observed polarization switching cannot be solely attributed to leakage-current contributions (Supplementary Fig. 12). To further evaluate the temporal stability of the switchable polarization, polarization retention measurements were performed using PUND pulse sequences with varied delay times. (Supplementary Fig. 13) The switchable polarization remains observable over the measured delay window, indicating that the field-induced polar state is not an instantaneous capacitive or leakage artifact. Meanwhile, the gradual decay of ΔP with increasing delay time suggests that the poled state is metastable and relaxes through thermally activated local dipolar reorientation within the amorphous matrix. This behavior is consistent with the relaxor-like nature of amorphous SA-Ph-Br, where polarization is governed by locally correlated dipolar configurations rather than long-range crystalline domains. Further humidity-dependent measurements reveal that the switchable polarization increases markedly with increasing humidity (Supplementary Fig. 14). This behavior suggests that water molecules modify the hydrogen-bonding environment of the salicylideneaniline proton-transfer units and plasticize the amorphous matrix, thereby facilitating field-induced local polar reorganization. Possible humidity-assisted protonic/ionic mobility or interfacial polarization may also contribute to the PUND-derived response^{41,42}.

The amorphous SA-Ph-Br film exhibits markedly faster photo-switching dynamics than its crystalline counterpart. Figure 5a compares the recovery from the UV-activated (vis-active) state to the ground (vis-inactive) state at room temperature. The amorphous film fully recovers within one minute, whereas the crystalline film requires over ten minutes under identical conditions. This dramatic acceleration arises from the disordered molecular network in the amorphous phase, which facilitates rapid proton-transfer relaxation and reduces diffusion barriers. The resulting ultrafast switching behavior renders amorphous SA-Ph-Br suggesting potential utility in adaptive optical and polarization-sensitive functionalities. The recovery dynamics are also thermally tunable. As shown by temperature-dependent in-situ CD and absorbance measurements (Fig. 3 and S6), lowering the temperature to 5°C extends the recovery time to over 50 min, demonstrating that molecular relaxation can be precisely modulated by the thermal environment. This tunability enables adaptive response speeds tailored to specific device requirements. Figure 5b further demonstrates a patterned amorphous film exhibiting reversible on–off switching under UV illumination, highlighting its potential for secure optical data storage and encryption.

The intrinsic chirality of SA-Ph-Br allows its physical properties to be modulated by the handedness of incident circularly polarized light (CPL). The photoresponse in our system includes UV-triggered enol-keto tautomerization and photoinduced carriers. Previous work has demonstrated that UV-triggered enol-keto tautomerization could tune the polarization by intrinsically tuning and reorientate molecular dipole. To investigate the effect generation of photoinduced carriers, we choose 405 nm to avoid triggering tautomerization. Figure 5c presents the CPL-dependent dielectric response of the S-amorphous SA-Ph-Br films under 405 nm. Upon UV irradiation of both LCP and RCP, the S-enantiomer exhibit an overall increase in dielectric permittivity thanks to the photoinduced carriers. Importantly, the dielectric response depends strongly on the light's helicity: LCP and RCP illumination yield distinct behaviors. For the S-amorphous film, LCP light produces a larger dielectric constant change, whereas RCP light gives a smaller response. The opposite trend is observed for the R-amorphous film (Supplementary Fig. 15), confirming that the interaction between CPL and molecular chirality governs the polarization dynamics through chiro-selective photoinduced carrier generation. This enantiomeric mirror symmetry corroborates the chiral origin of

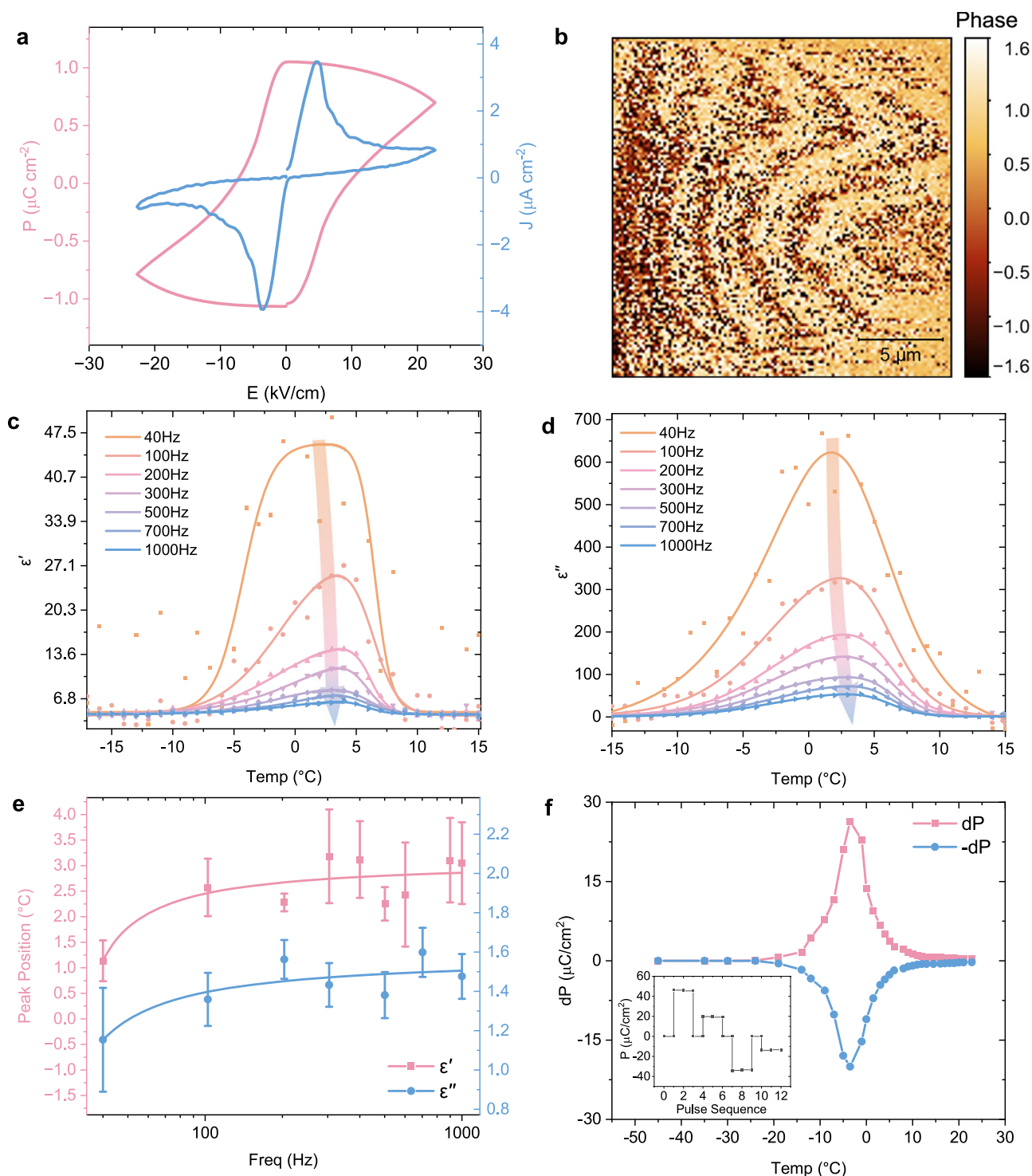


Fig. 4 | Relaxor-like ferroelectric response. **a** P-V loop of the S- and R-amorphous film. **b** PFM on R-amorphous film. **c, d** The dependence of real and imaginary part of dielectric permittivity on temperature showing Vogel-Fulcher relaxation.

e Extracted peak position and Vogel-Fulcher fitting of real and imaginary part of dielectric permittivity. **f** Extracted temperature-dependent switchable polarization from PUND measurements. Inset: PUND curve at 3.5 °C.

the optical control, demonstrating that the CPL can selectively modulate the thermal transport via the chiroptical excitation of photo-generated charge carriers. Furthermore, the photoresponse of ferroelectric behavior was investigated (Fig. 5c). Upon light irradiation, the saturation polarization decreases significantly, attributed to local heating and the injection of photoinduced carriers, which disrupt dipole-dipole ordering and reduce overall polarization. Importantly, the amplitude of the photoresponse depends on light helicity: for the

S-amorphous film, LCP illumination yields a larger reduction in saturation polarization than RCP. This mirror-symmetric dependence further demonstrates the chiroptical control of ferroelectric behavior through selective interaction between circularly polarized light and molecular chirality. This reversible coupling of structural disorder, ferroelectric relaxor behavior, and light-induced chirality establishes the foundation for reconfigurable chiral ferroelectrics. Building on these photoresponsive ferroelectric properties, we fabricated a

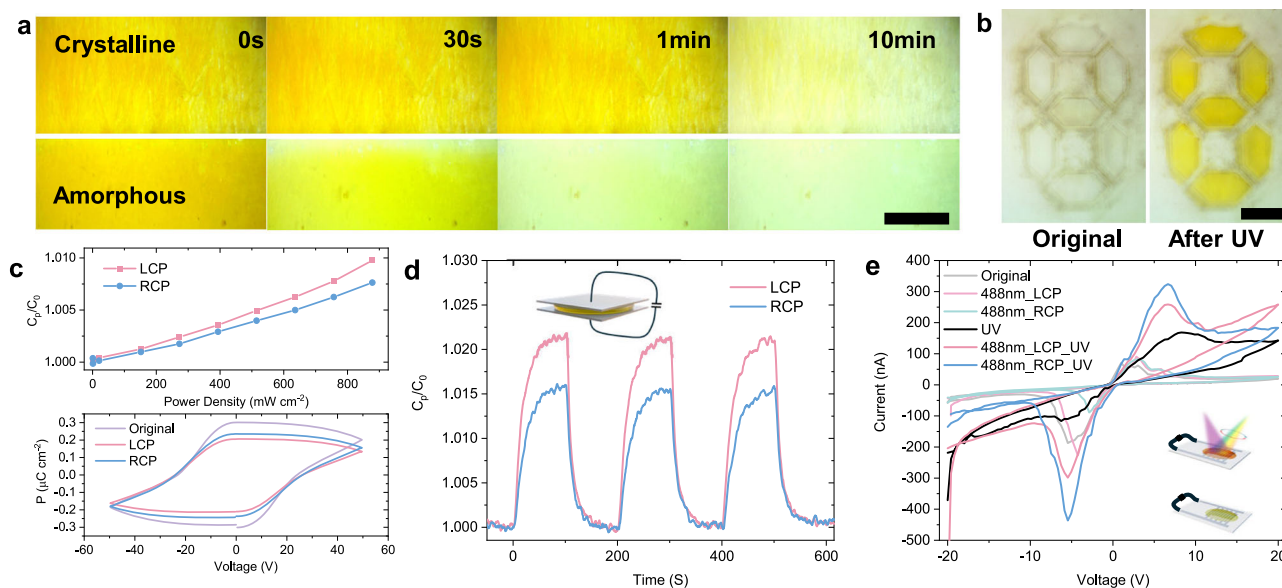


Fig. 5 | Proof-of-concept demonstration of light-modulated functional response. **a** Side-by-side comparison of crystalline and amorphous film on recovery from chiral to achiral state. Scale bar: 1 cm. **b** Printed device on-off switching under UV light. Scale bar: 1 cm. **c** 405 nm CPL effect on dielectric function and

ferroelectric P-E loop of S-amorphous sample. **d** capacitive CPL UV detector based on S-amorphous with 375 nm laser intensity of 113 mW cm⁻². **e** current-voltage UV-switchable visible CPL detector curves based on S-amorphous with 488 nm laser intensity of 250 mW cm⁻².

capacitive polarization-sensitive photodetector based on S-amorphous SA-Ph-Br (Fig. 5d). The device employs a sandwich configuration with top and bottom ITO electrodes enclosing the chiral amorphous ferroelectric layer. Upon 375 nm excitation, the device capacitance shows a power-dependent rapid increase followed by a fast decay upon light removal, confirming reversible photo-modulation of polarization. Notably, the response amplitude under the RCP illumination is substantially higher than under the LCP light, indicating intrinsic circular-polarization sensitivity. The opposite results are observed for the R-amorphous film. (Supplementary Fig. 16) To demonstrate the UV-switchable visible detection, we built a current-based photodetector (Fig. 5e). In the absence of UV excitation, the device remains inactive to visible light (488 nm). However, under continuous UV irradiation, it becomes photoactive in the visible range and retains distinct LCP/RCP-dependent responses, where the LCP illumination of visible light leads to substantially higher current than RCP illumination of visible light. Collectively, these results demonstrate a proof-of-concept UV-activated CPL-sensitive photodetector, effectively integrating molecular ferroelectricity with chiroptical switching for broadband and polarization-selective optical sensing. Moreover, thermal transport also displays handedness-dependent asymmetry for UV illumination (Supplementary Fig. 17): UV illumination enhances the thermal conductivity of the amorphous SA-Ph-Br, suggesting that photoinduced carrier generation promotes heat conduction. Importantly, for the S-amorphous film, LCP illumination leads to higher thermal conductivity than RCP, while the R-amorphous film shows the inverse relationship. These enantiomer-specific responses lay the groundwork for CPL-controlled thermal technologies.

Discussion

In summary, this work establishes amorphous SA-Ph-Br as a molecular platform in which structural disorder, photoinduced chirality, and field-responsive dipolar dynamics are intrinsically coupled. In contrast to previously reported photoresponsive ferroelectric materials, where light-driven polarization modulation is typically mediated by ordered crystalline lattices, amorphous SA-Ph-Br provides a distinct ferroic regime in which long-range crystallinity is removed while local molecular dipoles, photoactive proton-transfer units, and heterogeneous

dipolar environments are retained. The salicylideneaniline derivative SA-Ph-Br undergoes a UV-triggered achiral-to-chiral transition associated with enol-keto tautomerism and intramolecular proton transfer. After rapid vitrification, amorphous (S/R)-SA-Ph-Br exhibits a distinct ferroelectric hysteresis with a saturation polarization of -1 μC cm⁻² and a coercive field of -5.7 kV cm⁻¹, markedly lower than the -40 kV cm⁻¹ reported for its crystalline counterpart. The material displays frequency-dependent dielectric relaxation (40 Hz–1 kHz) and a maximum switchable polarization of -26 μC cm⁻² near -3.5 °C, consistent with relaxor-like dipolar dynamics arising from structurally heterogeneous local environments.

Importantly, the amorphous phase exhibits markedly faster and more tunable photo-switching kinetics than the crystalline state. The UV-activated chiral state recovers to the achiral state within ~1 min, over an order of magnitude faster than in the crystalline phase, and tunable to >50 min at 5 °C, enabling precise control of photo-switching kinetics. Furthermore, the circularly polarized light-dependent dielectric and thermal responses show mirror-symmetric behavior between (S)- and (R)-enantiomers, confirming chiro-selective excitation and polarization coupling. A UV-activated, helicity-sensitive capacitive photodetector based on amorphous (S)-SA-Ph-Br demonstrates power-dependent capacitance modulation under 375 nm illumination, providing a proof-of-concept demonstration of coupled chiroptical and polarization response. Together, these results differentiate amorphous SA-Ph-Br from crystalline photoresponsive ferroelectrics by showing that molecular vitrification can serve as an active design strategy for disorder-enabled ferroic functionality. More broadly, amorphous molecular solids offer a previously under-explored platform for integrating photoinduced chirality, local dipolar disorder, and reconfigurable polarization dynamics beyond crystalline order.

Methods

Chemicals and sample preparation

3,5-di-tert-butyl-2-hydroxybenzaldehyde (99%), (R)-(+)-1-(4-Bromophenyl)ethylamine (≥96.0%) and (S)-(–)-1-(4-Bromophenyl)ethylamine (≥98.0%) were purchased from Sigma-Aldrich. All chemicals were used as received without further purification.

The SA-Ph-Br crystals with S- and R- handedness were prepared following the previous literature²². Briefly, 3,5-di-tert-butyl-2-hydroxybenzaldehyde and (S)- or (R)- 1-(4-bromophenyl)ethan-1-amine were firstly dissolved in methanol and the mixture was stirred at 68 °C overnight. The (S)- or (R)- crystals with yellow color are collected with slow evaporation of the solution. To prepare (S/R)- amorphous solid, the (S/R)-crystal was first heated up to 110 °C on hot plate to acquire the liquid and was then rapidly cooled to room temperature.

Measurements

SEM images of the crystalline and amorphous films were acquired on Tescan XEIA FEG SEM in AIMLab in the Maryland NanoCenter. AFM image was acquired with Bruker Dimension 3100 system. Conductive Pt/Ir-coated silicon probes (EFM, Nanoworld) with a nominal spring constant of ≈ 2.8 nN nm⁻¹ and a resonance frequency of ≈ 75 kHz were used for piezoelectric domain imaging. CD spectra and in-situ CD measurement are done on Jasco J-810 spectropolarimeter equipped with PFD-425S temperature control unit. Continuous-wave lasers with wavelengths of 375 nm, 405 nm, 488 nm lasers were used for light-related measurements. The laser beam was first passed through a linear polarizer to generate a well-defined linearly polarized light, which was then passed through a rotatable quarter-wave plate. By adjusting the fast axis orientation of the rotatable quarter-wave plate, LCP or RCP can be generated. The resulting CPL was directed perpendicularly onto the surface of the device under test. Under identical illumination intensity, the dielectric/thermal conductivity/I-V responses of the device to LCP and RCP light were measured. Temperature-dependent measurement was performed with a box furnace or cryostat. The CPL-dependent and temperature-dependent P-V hysteresis loop and PUND measurements were measured using a Precision LC ferroelectric tester. Humidity-dependent P-V loops were conducted by putting the samples into a container where humidity was controlled by a humidifier. CPL-dependent and temperature-dependent dielectric function and conductance were measured with Agilent 4294 A precision impedance analyzer. The CPL-dependent and temperature-dependent I-V curve was recorded using a Keithley 2450 sourcemeter. Thermal analysis was measured using Perkin Elmer DSC 7. Powder X-ray diffraction (PXRD) data were measured using a Bruker D8 Advance X-ray diffraction system with Cu K α radiation. Raman spectra of the sample were measured using HORIBA Jobin Yvon Raman microscope in UMD Surface Analysis Center. Fourier-transform infrared spectroscopy (FTIR) spectra of the sample were measured on an Agilent Cary 630 FTIR spectrometer.

Data availability

The data supporting the findings of this study are available within the article and its Supplementary Information files. Source data are available with the paper. All data are available from the corresponding author upon request. Source data are provided with this paper.

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Author contributions

H.L. and Y.X. prepared the materials. H.L. performed the experiments and analyzed the results. T.X. performed piezoresponse force microscopy under the supervision of C.G., S.R., and H.L. conceived the idea. H.L. wrote the manuscript. All authors discussed the results and reviewed the manuscript.

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Correspondence and requests for materials should be addressed to Hanyu Liu or Shenqiang Ren.

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